

# ClimateLife - Smart Pot

## Session1 - Group11

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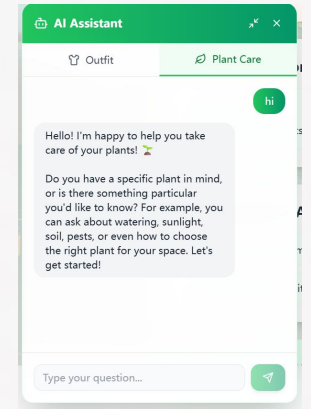
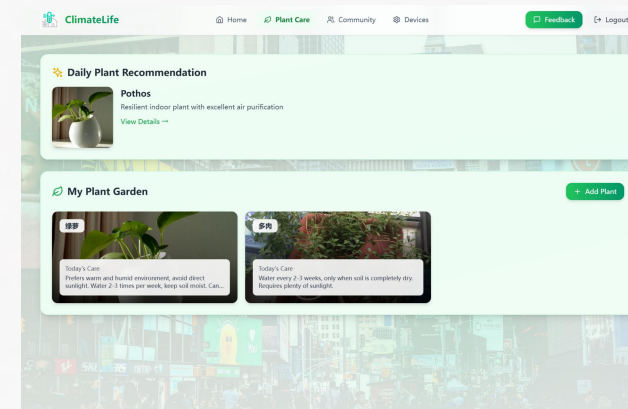
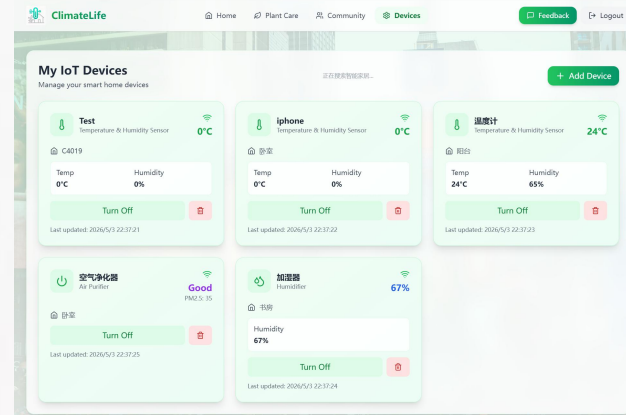
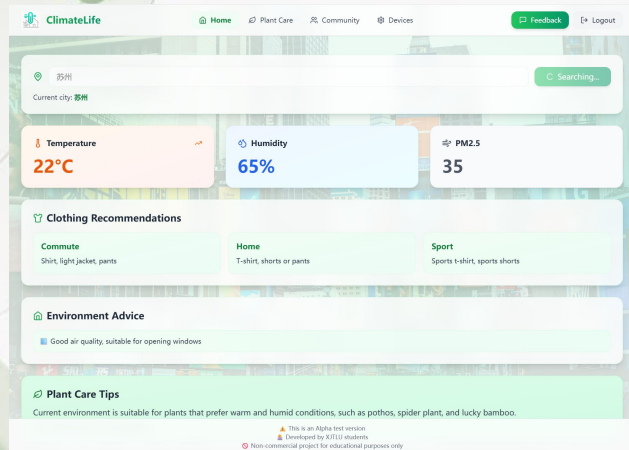
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Sirui Zeng, Zhihan Wang, Zehao Wu

From a smart living website to a physical IoT plant-care system.



# Why we built ClimateLife Smart Pot ?

**Background :** Last semester, we built the ClimateLife website.



## Location & Weather

- Check real-time temperature, humidity, and air quality in your city

## Add IoT Devices

- Connect sensors to display real-time environmental data

## Add Your Plants & AI Assistant

- Users can add and manage their personal plants and growing records
- AI assistant provides care guidance and recommendations

# Why we built ClimateLife Smart Pot ?

## Smart Flowerpot Product Concept

Automatic Watering | Smart Supplementary Lighting | Environmental Monitoring | Web Connectivity



Smarter plant care,  
better everyday life



Intelligent care for  
healthy growth



Cloud-connected control,  
anytime and anywhere

- **Problem Detected** : Users want more **AI** and **IoT** support

5.您希望未来增加哪些功能?What new features would you like to see added?

|                                             |    |                        |        |
|---------------------------------------------|----|------------------------|--------|
| A.AI个性化助手 / AI Personal Assistant           | 21 | <div><div></div></div> | 77.78% |
| E.更多IoT设备兼容 / More IoT Device Compatibility | 12 | <div><div></div></div> | 44.44% |

- **Pain Point** : Busy **students** and young **professionals** often forget to water plants and don't know plant conditions

2.您的职业类别是?What is your occupation? [Single Choice]

| Option #          | Subtotal # | Percentage                    |
|-------------------|------------|-------------------------------|
| A.学生 / Student    | 19         | <div><div></div></div> 70.37% |
| B.在职人员 / Employed | 3          | <div><div></div></div> 11.11% |

1您的年龄段是? What is your age range? [Single Choice]

| Option #           | Subtotal # | Percentage                    |
|--------------------|------------|-------------------------------|
| A.18岁以下 / Under 18 | 3          | <div><div></div></div> 11.11% |
| B.18—25            | 16         | <div><div></div></div> 59.26% |



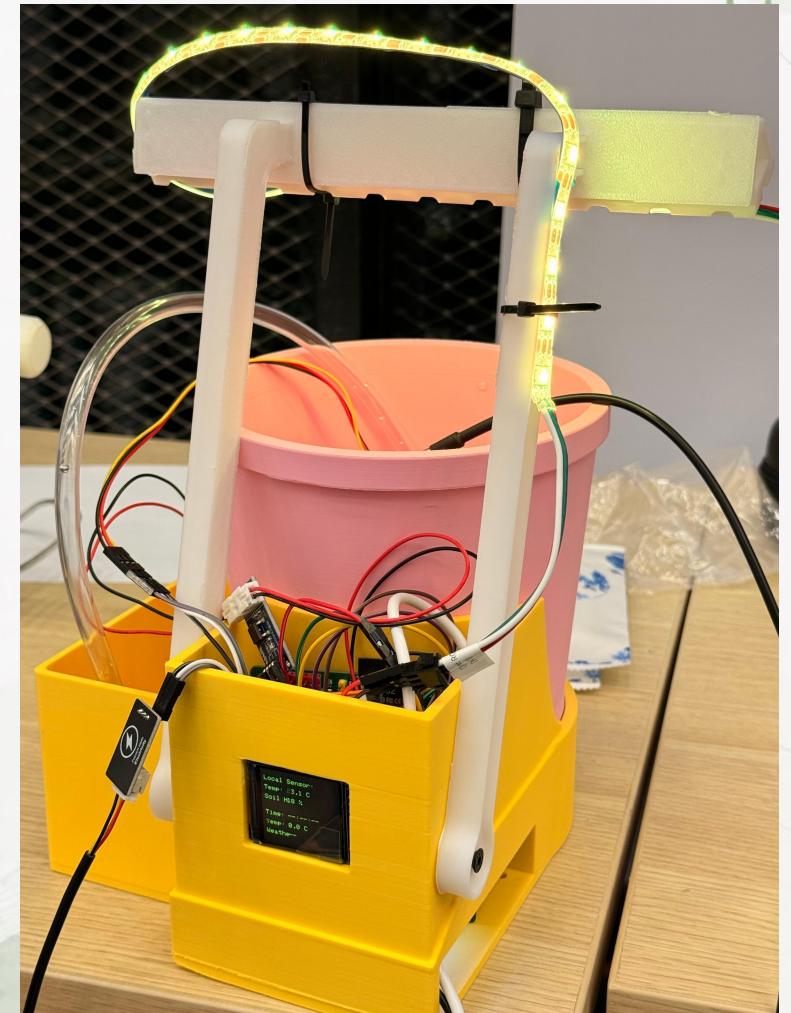
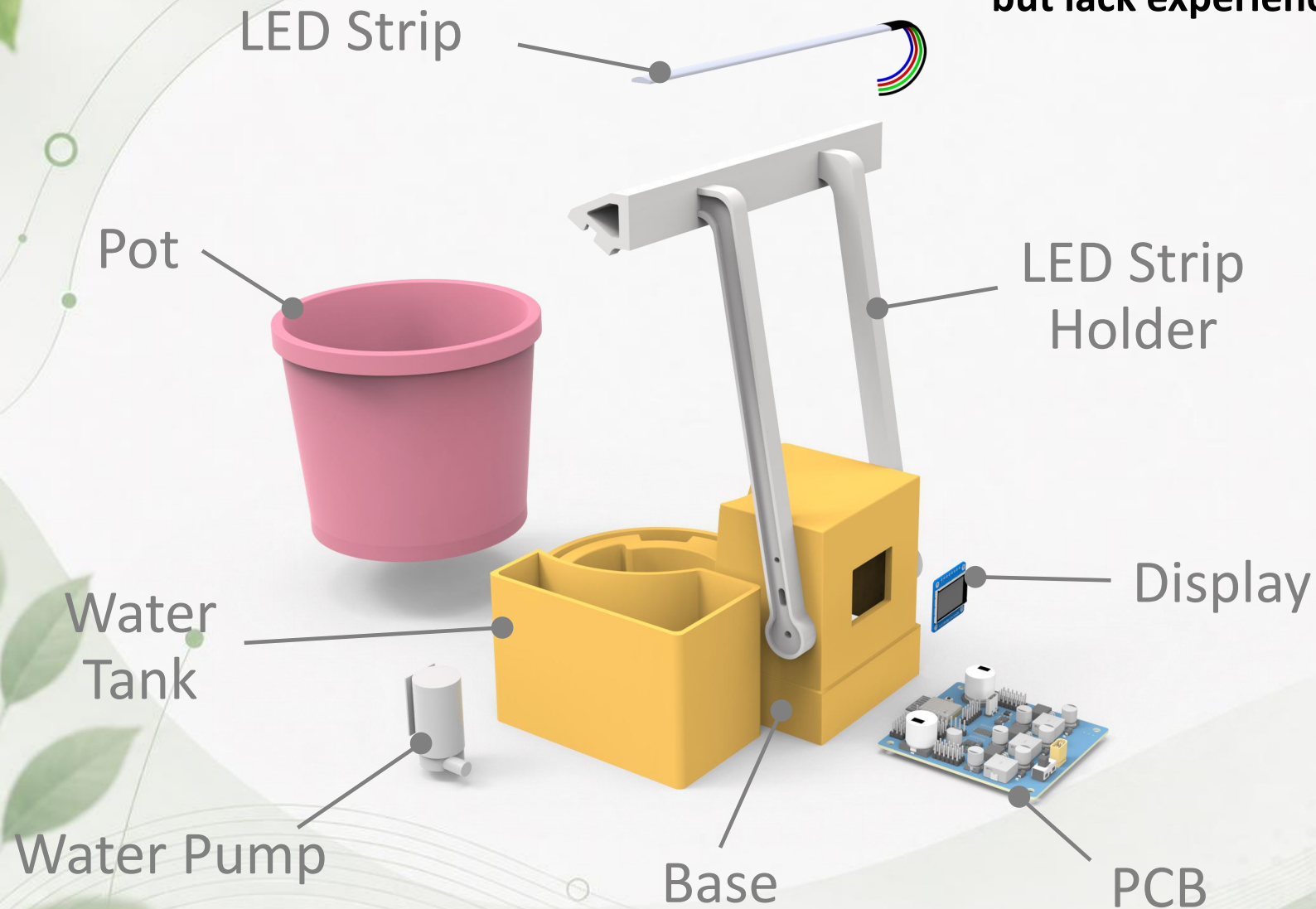
- **Conclusion** : The website can to be upgraded to a **physical smart pot system** with real-time data and controllable features

“Why is there so much water???”



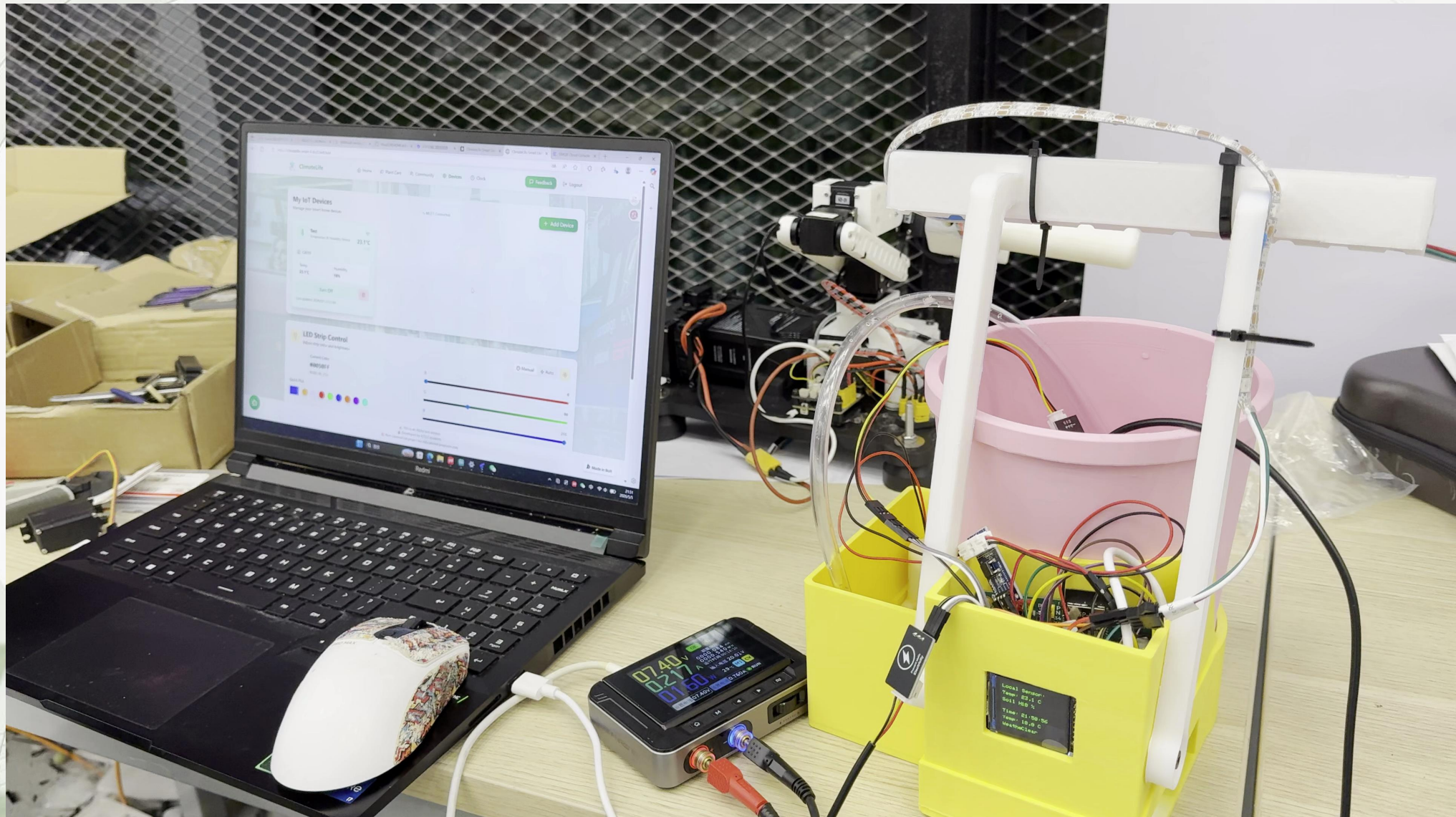
# What We Built ?

For students and young professionals who enjoy plants but lack experience or time to care for them.





# Live Demo



# User Validation - Method



## Target Users

- Busy students and young professionals
- **20 users**



## Testing Approach

- Longitudinal study
- Create user profiles
- Each user participated in **3 rounds**  
(week4, week6, week8)



## Goal

- Observe user interactions and gather feedback to improve the product
- Not only collecting opinions but also performing usability tests



# User Validation - Iteration Process

## Iteration 1 – Concept Validation (week4)

- Users understood the smart pot idea
- They wanted real-time environmental data first



## Prototype V1

- Built the first pot prototype
- Connected temperature and humidity data to the website

|               |        |          |
|---------------|--------|----------|
| Local Sensor: | Temp   | Humidity |
| Temp: 24.7 C  | 25.2°C | 42%      |
| Soil H42 %    |        |          |

## Iteration 2 – Data Usability Test (week6)

- Users could see data
- Users wanted more features, e.g., controlling lighting and watering



## Prototype V2

- Added web-controlled grow light system
- Enabled users to control lighting through the system



### LED Strip Control

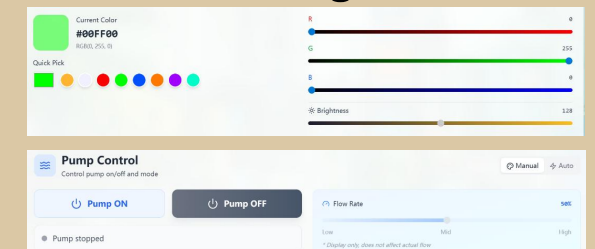
Adjust strip color and brightness

## Iteration 3 – Light Control Test (week8)

- Users could control the light successfully
- Users suggested different plants might need different light settings and wanted watering functionality

## Prototype V3

- Added customizable light interface
- Added watering function



# What Comes Next

## Limitation:

- All the functions depends on stable network connection

## Next Steps:

### More functions

- Integrate water pump fully and optimize control interface
- Add more plant types, set optimal light & watering mode for different plant
- Implement automatic control of light and watering based on temperature & humidity

### AI assistant

- Improve AI assistant guidance with more tips and reminders

### User Testing

- Prepare for long-term user testing and feedback collection





# Thank you

